

Highlights

- Attribute-decomposed attention gives a large observed gain on COGS ($d=2.12$).
- Cyclic head assignment is characterized as one member of an equivalent family.
- A pairing-strategy ablation shows random-fixed assignment matches cyclic.
- Probing finds no evidence of emergent attribute-slot specialization.
- All results, including null findings, are reported with full seed-level data.

Attribute-Decomposed Attention: A Relational Inductive Bias for Structured Reasoning

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Abstract

We introduce **Attribute-Decomposed Attention** (RelAttn), an attention mechanism that decomposes token representations into k typed attribute slots and restricts each head to a single attribute pair, a neural analogue of a foreign-key join predicate. We prove subsumption of standard dot-product attention ($k=1$), characterize valid head-assignment strategies as directed Hamiltonian cycles on the attribute slots (of which the cyclic assignment used throughout is one canonical member, not a unique optimum), and establish gradient-isolation properties.

We report a task-dependent behavioral study across three benchmarks, each with a dedicated ablation, with full seed-level statistics throughout. On **compositional generalization** (COGS), RelTransformer reaches $5.81\% \pm 1.34$ generalization-split exact match vs. $2.27\% \pm 1.94$ for Standard Transformer (3 seeds, Cohen’s $d=2.12$, one-tailed $p=0.034$), with far greater cross-seed stability; this is the clearest evidence for the architecture’s benefit, and we treat it as an observed effect requiring further replication rather than a settled result given the small seed count. On **mathematical reasoning** (GSM8K, from-scratch), an 8-seed comparison gives $2.79\% \pm 0.19$ vs. $2.54\% \pm 0.31$ ($d=0.96$, one-tailed $p=0.040$,

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two-tailed $p=0.079$); inspecting predictions reveals substantial answer-frequency mode collapse in both architectures, so we treat this as a modest, secondary behavioral difference rather than evidence of improved reasoning. On **text-to-SQL** (Spider), the two architectures are statistically indistinguishable once both have a copy mechanism, indicating that schema-linking rather than the attention mechanism dominates this task. A pairing-strategy ablation is consistent with the theoretical characterization of head-assignment strategies below: a random-fixed key-side assignment matches cyclic ($d=0.06$, n.s.), providing no evidence that the canonical cyclic assignment is uniquely advantageous. We additionally probe for the role-specific slot specialization the design might suggest – using the model’s real training task, content-based role labels, and a permutation-null control – and find no discriminability signal above an untrained, randomly initialized model: structural decomposition delivers the behavioral effects above without producing emergent, inspectable specialization at the representation level. All results, including nulls, are reproducible from the released code and logs at <https://github.com/muxiddin19/relational-attention>.

Keywords: text-to-SQL, relational attention, functional dependencies, compositional generalization, mechanistic interpretability, reproducibility

1. Introduction

Natural language interfaces to databases converting user questions into SQL queries represent a central challenge in data engineering [1, 2, 3]. Generating accurate SQL requires compositional reasoning: understanding how table columns
5 relate to each other, how foreign-key constraints determine join predicates, and how database schemas structure entity relationships. Despite dramatic progress from pretrained language models [4, 5], the *attention mechanism* at the heart of these models was designed for general-purpose pattern matching, not for relational reasoning.

10 Standard dot-product attention [6] computes token similarity over entire

d -dimensional vectors:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V \quad (1)$$

This formulation conflates all semantic aspects of a token its entity identity, its role in a join predicate, its table membership into a single similarity score. For database-oriented tasks, where the distinction between *which attribute is being*
 15 *compared* matters structurally, this is a fundamental mismatch.

Motivating Example. Consider the question “Find all students enrolled in both Databases and Algorithms”. The correct SQL requires a self-join:

```
SELECT s.name FROM student s
      JOIN enrollment e1 ON s.id = e1.sid
20    JOIN enrollment e2 ON s.id = e2.sid
WHERE e1.course='DB' AND e2.course='ALG'
```

The token `s.id` participates in three structurally distinct roles simultaneously:
 a) as an entity identifier selecting from `student`; b) as the join key in the first
 enrollment predicate; and c) as the join key in the second enrollment predicate.
 25 Standard attention collapses all three into one similarity score. Attribute-
 Decomposed Attention assigns each role to a dedicated attribute slot – slot 1 for
 entity identity, 2 for the first join predicate, 3 for the second – and computes
 attention across the appropriate slot pairs. This structural alignment reduces
 the multi-join task to a set of independent single-attribute comparisons, each
 30 semantically grounded in the schema.

We propose Attribute-Decomposed Attention, which introduces a relational
 inductive bias by decomposing token representations into k typed attribute slots.
 The key insight, inspired by relational schema theory, is that each attention head
 should perform a specific attribute-to-attribute comparison — a neural analogue
 35 of a join predicate. A head matching slot j of query tokens against slot l of
 key tokens is motivated by the functional dependency $\mathbf{a}^{(j)} \rightarrow \mathbf{a}^{(l)}$, capturing the
 same structural relationship that foreign-key constraints encode in relational

databases. We use this correspondence throughout as an *FD-inspired inductive bias* guiding the head-assignment design, not as a claim that a trained head provably implements FD semantics in Codd’s formal sense (Sec. 4 states this precisely); a dot-product similarity between learned embeddings is a soft, learned association, not a relational join. Our contributions are:

1. A mechanism decomposing tokens into k attribute slots with join-pair heads, attribute gating, and attribute mixing, grounded in relational model theory. Formal results establish: a) balance, path-completeness, and minimum-edge together characterize head-assignment strategies as directed Hamiltonian cycles on the k attribute slots, of which the cyclic pattern is the canonical instance used throughout (Theorem 4); b) standard attention aggregates all k diagonal attribute-pair terms into one undifferentiated score with no access to off-diagonal pairs, while Join Attention isolates a single named pair as an independently parameterized, inspectable unit (Proposition 2); c) gradient isolation at attribute embedding level (Proposition 7); d) $O(n^2d)$ complexity.
2. **A task-dependent behavioral study with full statistical reporting.** On COGS (with copy mechanism), RelTransformer’s generalization-split EM is more than double Standard Transformer’s ($5.81\% \pm 1.34$ vs. $2.27\% \pm 1.94$, Cohen’s $d=2.12$, one-tailed $p=0.034$, 3 seeds) with far greater cross-seed stability; we treat this as the strongest evidence in the paper while noting that $n=3$ warrants replication. On GSM8K, an 8-seed comparison shows a small difference ($2.79\% \pm 0.19$ vs. $2.54\% \pm 0.31$, $d=0.96$, one-tailed $p=0.040$, two-tailed $p=0.079$) that inspection of predictions attributes largely to answer-frequency mode collapse rather than improved reasoning. On Spider (with copy mechanism), the two architectures are statistically indistinguishable, indicating schema-linking rather than attention architecture dominates text-to-SQL performance at this scale. Component ablations show gating and mixing each contribute (removing either measurably hurts GSM8K accuracy), and a pairing-strategy ablation (Sec. 6)

shows a random-fixed key-side assignment matches cyclic ($d=0.06$, statistically indistinguishable) – evidence that the canonical cyclic assignment is not uniquely advantageous, consistent with (though not a confirmation of) Theorem 4’s characterization of a family of valid assignments.

3. **A negative interpretability finding.** We probed the trained model for the role-specific attribute-slot specialization the architecture’s design might suggest, using the model’s actual training task, content-based role labels, and a permutation-null control (Sec. 7). We find no discriminability signal distinguishable from an untrained, randomly initialized model of identical architecture: structural decomposition delivers the behavioral effects above without producing emergent, inspectable specialization at the representation level, a finding we believe is useful to the interpretability community as a caution against assuming architecture-induced interpretability.

2. Related Work

RAT-SQL [7] uses relation-aware self-attention encoding schema relations as edge types. BRIDGE [8] uses schema-linking with a hybrid representation of questions and schemas. Both rely on copy mechanisms and schema-linking components that are the dominant factors for SQL generation accuracy; our experiments confirm this: without these components, both RelTransformer and Standard Transformer achieve 0% EX on Spider from scratch, consistent with the schema-linking hypothesis.

More recent systems leverage pretraining: RASAT [4] integrates relational structures into T5, and SQLformer [5] uses autoregressive graph generation over T5-Large. These pretrained systems (trained on ≥ 1 B tokens) provide context but are not direct competitors to our from-scratch models.

Among system-level NL2SQL approaches, ValueNet [9] augments SQL generation with database value information; GAR [3] uses a generate-and-rank pipeline; and CatSQL [1] employs sketch-based templates with semantic correction. The comprehensive NL2SQL360 benchmark [2] evaluates 20 NL2SQL

systems, highlighting that structural generalization remains a key open challenge. Our approach addresses this at the attention-mechanism level rather than through system-level engineering.

100 The relational model of Codd [10] defines data as sets of tuples over a schema $\mathcal{R} = \{A_1, A_2, \dots, A_m\}$. A *functional dependency* (FD) $X \rightarrow Y$ holds in relation R if for all tuples $t_1, t_2 \in R$: $t_1[X] = t_2[X] \Rightarrow t_1[Y] = t_2[Y]$. FDs form the basis of normalization and are the structural commitments encoded in foreign-key constraints. The key operation in SQL the *equi-join* $R \bowtie_{A=B} S$ assembles tuples
 105 sharing the same value on attribute A in R and attribute B in S , precisely the pattern encoded by Join Attention when query-slot j encodes A -type information and key-slot l encodes B -type information.

Attention mechanisms have been connected to database operations before in spirit but not in formal treatment. Transformer cross-attention $\text{Attn}(Q, K, V)$
 110 is interpretable as a soft lookup of key-value store (K, V) using queries Q ; however, this analogy treats the entire key vector as the “lookup key” without decomposition into typed attributes. Our contribution is to show that restricting each head to a specific attribute pair formally isolating one FD direction yields both better empirical performance and a principled theoretical connection to the
 115 relational model. This is directly analogous to the move from full-table scans to indexed lookups, a well-understood efficiency-and-expressiveness trade-off in relational databases.

The database community has increasingly applied attention and learned representations to core query problems. Kumar et al. [11] survey the founda-
 120 tional data management challenges in ML systems. ALECE [12] shows that attention over join statistics achieves SOTA cardinality estimation on dynamic SPJ workloads; PRICE [13] extends this with cross-database pretrained transfer. Li et al. [14] apply deep cost modeling to resource-aware query optimization. Picado et al. [15] develop scalable relational learning under integrity constraints.
 125 Rissaki et al. [16] connect database view semantics to neural representation learning. Our work shows that grounding the attention *mechanism itself* in functional-dependency theory yields systematic gains across structured reasoning

tasks.

Grouped Query Attention (GQA) [17] shares key-value heads across query
130 groups for efficiency. Perceiver IO [18] uses latent bottleneck cross-attention.
Both optimize efficiency, while ours expressiveness for structured tasks. Relative
position attention [19] injects positional structure via bias terms; our attribute
decomposition operates on the content dimension.

A note on naming. Diao and Loynd [20] also use the name “Relational At-
135 tention” for a distinct mechanism: they associate a vector state with every *edge*
(token pair) and update it via a triangular attention operation composing two ad-
jacent edges, evaluated on graph-structured tasks including the CLRS algorithmic
reasoning benchmark. Our RelAttn instead decomposes each individual *token*
into k typed attribute slots and restricts each head to a single attribute-pair com-
140 parison; we maintain no explicit per-pair edge state and perform no cross-edge
composition within a layer. The two mechanisms are complementary rather than
competing: theirs generalizes attention to operate over pairs as first-class objects,
ours restricts which sub-vectors of each token a given head may compare. We
flag the naming collision explicitly to avoid confusion and note it as a candidate
145 architectural composition for future work — e.g., attribute-decomposed *edge*
states, which their triangular update mechanism (Sec. 8) suggests as a natural
way to add explicit single-layer relational composition, which our current cyclic
pairing achieves only across layers.

Positioning in one place. Three architectures inject relational structure
150 into attention in genuinely different ways, and distinguishing them precisely
matters for evaluating novelty: RAT-SQL [4] injects *explicit* relational structure
by adding schema-relation-type embeddings as bias terms on top of standard
dot-product attention, requiring a pre-defined relation schema; Diao and Loynd’s
Relational Attention [20] associates an explicit vector state with every token-pair
155 *edge* and updates it via triangular composition of adjacent edges; RelAttn instead
restricts *which sub-vectors of each token* a given head may compare, requiring
no pre-defined relation schema and no explicit edge state. The first two add
machinery on top of or alongside standard attention; RelAttn instead constrains

standard attention’s own computation at the representation level.

160 Several parallel lines decompose representations for structured reasoning. Slot Attention [21] iteratively routes inputs to learned slots via competitive softmax for object-centric decomposition; RelAttn differs in using fixed per-head slot assignments learned via task gradients. Set Transformer [22] uses learnable inducing points for permutation-invariant efficiency; its goal is compute 165 reduction, not relational expressiveness. Neural Logic Machines [23] implement differentiable logic over tensors indexed by object tuples, requiring explicit object segmentation; RelAttn operates on token sequences directly. Tensor Product Representations [24] bind roles to fillers via outer products for symbolic compositionality; attribute slots approximate this role-filler binding end-to- 170 end without pre-specified roles. Grammar-constrained decoding [25] enforces syntactic validity at inference, which we adopt uniformly. RelAttn is distinctive in its grounding in relational database theory FDs, FK-join predicates, normal forms and application to typed schemas.

Two architectural families are conceptually closest and should be discussed. 175 Dual-attention approaches (e.g., Dual Attention Transformer, DAT) explicitly disentangle sensory from relational processing via parallel attention streams, demonstrating consistent gains on symbolic reasoning. RelAttn differs in that all streams operate over the same token sequence (no pre-segmented relational vs. sensory channels) and the decomposition follows relational model theory directly — each head computes a typed attribute-pair join predicate rather than a channel- 180 level mixture. DAT and FOLNet codebases exist in vision/general-language research repositories, but no implementation compatible with our autoregressive encoder-decoder setup (teacher-forced seq2seq, 32K-vocab, FP32) is publicly available; a controlled empirical comparison is an important future direction. 185 First-order logic attention networks (e.g., FOLNet) frame attention as soft first-order logic, providing universal operators that subsume standard attention; RelAttn’s join-pair formulation is a restricted class of these relational operators with deterministic, theory-motivated slot-role assignments and FD-grounded cyclic coverage. Both approaches are evaluated primarily in general language or

190 vision settings; our contribution is to show that grounding the decomposition in
functional-dependency theory yields measurable gains in structured reasoning
tasks over typed relational schemas.

Recent dual-stream architectures separate sensory and relational attention
channels in vision models, or compute explicit relation tensors over pre-segmented
195 object pairs for visual reasoning. RelAttn differs in operating on token sequences
without pre-segmented objects, and grounds the decomposition in relational
model, typed database schemas.

The emergence of billion-parameter language models has driven a paradigm
shift in NL2SQL research. PICARD [25] introduces constrained auto-regressive
200 decoding that filters invalid SQL tokens at each step using an incremental parser;
applied to T5-Large, it achieves 75.5% EX on Spider dev without schema-linking
preprocessing. RESDSQL [26] decouples schema linking from SQL skeleton
parsing into a two-stage pipeline; with T5-3B (3 billion parameters), it achieves
79.9% EX. DIN-SQL [27] decomposes SQL generation into modular sub-problems
205 solved in-context by GPT-4, reaching 82.8% EX. The systematic benchmark by
Gao et al. [28] evaluates ten LLM-based systems on Spider and BIRD, finding
GPT-4 achieves 86.6% EX on Spider with optimal prompting.

These approaches represent a fundamentally different regime from ours: they
rely on large-scale pretraining corpora alongside task-specific fine-tuning. Our
210 work asks a complementary question: *given a fixed model capacity and training
budget, what structural inductive bias in the attention mechanism most benefits
relational reasoning?* This matters for deployment scenarios where LLM APIs
are unavailable (on-premise databases, privacy-sensitive applications, edge de-
ployments) and for understanding which aspects of structured reasoning arise
215 from architecture rather than scale. Separately listed for landscape context:
pretrained systems include PICARD T5-Large (75.5%, 770M, ≥ 1 B token pre-
training) and RESDSQL T5-3B (79.9%) in Table 2; these are *not* comparable
to from-scratch rows due to the pretraining information advantage, and no
performance claims are made across this regime boundary.

220 Several systems encode the database schema as a graph and use graph neural

networks (GNNs) or graph-attention to propagate structural information. RAT-SQL [7] encodes 33 relation types between question tokens, table tokens, and column tokens as edge embeddings in a relation-aware transformer. Ours re-implements RAT-SQL without the T5 pretraining used in the original, following
225 standard practice for architectural comparisons. RelTransformer and RAT-SQL schema encoding are designed to be complementary: explicit schema edge types from RAT-SQL address known structural priors while RelAttn handles implicit relational dependencies. Their combination is a key architectural hypothesis for NL2SQL data engineering applications, motivating future work on combined
230 schema-linking and relational-attention architectures.

Standard transformers struggle with systematic generalization [29]. We evaluate on COGS [30], SCAN [31, 32], and CFQ [33]. Pretrained T5-Base baselines on these benchmarks are from [34]. Ruis et al. [32] introduce a comprehensive grounded-language benchmark for systematic generalization, providing a richer
235 evaluation setting beyond the original SCAN.

3. Attribute-Decomposed Attention

Definition 1 (Attribute Decomposition). *Given token representation $\mathbf{x} \in \mathbb{R}^d$, decompose into k attribute embeddings:*

$$\mathbf{x} = [\mathbf{a}_1; \mathbf{a}_2; \dots; \mathbf{a}_k], \quad \mathbf{a}_i \in \mathbb{R}^{d/k} \quad (2)$$

where $[\cdot; \cdot]$ denotes concatenation and d is divisible by k .

240 Attributes are learned via k separate linear projections:

$$\mathbf{a}_i^{(j)} = W_j^{(e)} \mathbf{x}_i, \quad W_j^{(e)} \in \mathbb{R}^{(d/k) \times d} \quad (3)$$

Each projection is followed by layer normalization. No supervision guides attribute semantics; whether interpretable specialization emerges through task gradients is an empirical question we test directly and report honestly in Sec. 7 (we do not find evidence that it does, though the architecture still yields real
245 behavioral gains).

Definition 2. For query sequence $Q \in \mathbb{R}^{n \times d}$ and key sequence $K \in \mathbb{R}^{m \times d}$, with join on query attribute j and key attribute l :

$$\alpha_{it} = \frac{\exp(\mathbf{a}_i^{(j)} \cdot \mathbf{b}_t^{(l)} / \tau)}{\sum_{s=1}^m \exp(\mathbf{a}_i^{(j)} \cdot \mathbf{b}_s^{(l)} / \tau)} \quad (4)$$

where $\mathbf{a}_i^{(j)} \in \mathbb{R}^{d/k}$ is attribute j of query token i , $\mathbf{b}_t^{(l)} \in \mathbb{R}^{d/k}$ is attribute l of key token t , and $\tau = \sqrt{d/k}$.

250 3.1. Attribute Gating and Mixing

After aggregating values ($\mathbf{h} = \alpha V$), we apply content-conditional gating:

$$\text{Gate}_{f_\phi}^{(j)}(\mathbf{h}) = \sigma(f_\phi(\mathbf{a}^{(j)})) \cdot \mathbf{h} \quad (5)$$

f_ϕ is a two-layer MLP with sigmoid output. This implements content-dependent on/off routing based on attribute j 's value. The gated output $\mathbf{g} = \text{Gate}^{(j)}(\mathbf{h})$ is then recombined across all k attribute slots. $\mathbf{g}_i$ for the i -th slot's contribution

255 from $\mathbf{g}$:

$$\text{Mix}_{\mathbf{w}}(\mathbf{g}) = \sum_{i=1}^k \text{softmax}(\mathbf{w})_i \cdot W_i \mathbf{g}_i \quad (6)$$

The W_i are per-slot learned projections; Mix operates on the gated slot values, not the original attribute embeddings $\mathbf{a}_i$.

Definition 3 (Relational Attention). For query Q , key K , value $V \in \mathbb{R}^{n \times d}$ and attribute pair (j, l) :

$$\alpha = \text{JoinAttn}^{(j,l)}(Q, K), \quad \mathbf{h} = \alpha V \quad (\text{value aggregation})$$

$$\mathbf{g} = \text{Gate}^{(j)}(\mathbf{h}) \quad (\text{slot gating})$$

$$\tilde{\mathbf{h}} = \text{Mix}_{\mathbf{w}}(\mathbf{g}) \quad (\text{slot mixing})$$

$$\text{output} = W_O \tilde{\mathbf{h}} \quad (\text{projection})$$

260 For multi-head attention with h heads, head r uses cyclic pairing $(j_r, l_r) = (r \bmod k, (r+1) \bmod k)$. Outputs are concatenated and projected:

$$\text{RelAttn}(Q, K, V) = \text{Concat}(\text{head}_0, \dots, \text{head}_{h-1}) W_O. \quad (7)$$

The cyclic assignment ensures every attribute index appears as query-source in exactly one head and as key-source in exactly one head, providing complete coverage of all k adjacent attribute-pair transitions per layer. For $h > k$, the cyclic
 265 pattern wraps: heads r and $r+k$ share pair $(r \bmod k, (r+1) \bmod k)$ but maintain *independent* value matrices W^V , allowing them to attend to different contexts for the same relational role. With $h=mk$, each pair receives $m \times$ coverage per layer.

The multi-head RelAttn layer has time complexity $O(n^2 d)$ matching stan-
 270 dard multi-head attention because each of the h heads computes an $(n \times d/k)$ -dimensional attention map ($O(n^2 d/k)$ per head, $h = k$ heads total, giving $O(n^2 d)$). Space complexity is $O(n^2 + nd)$, also matching standard attention.

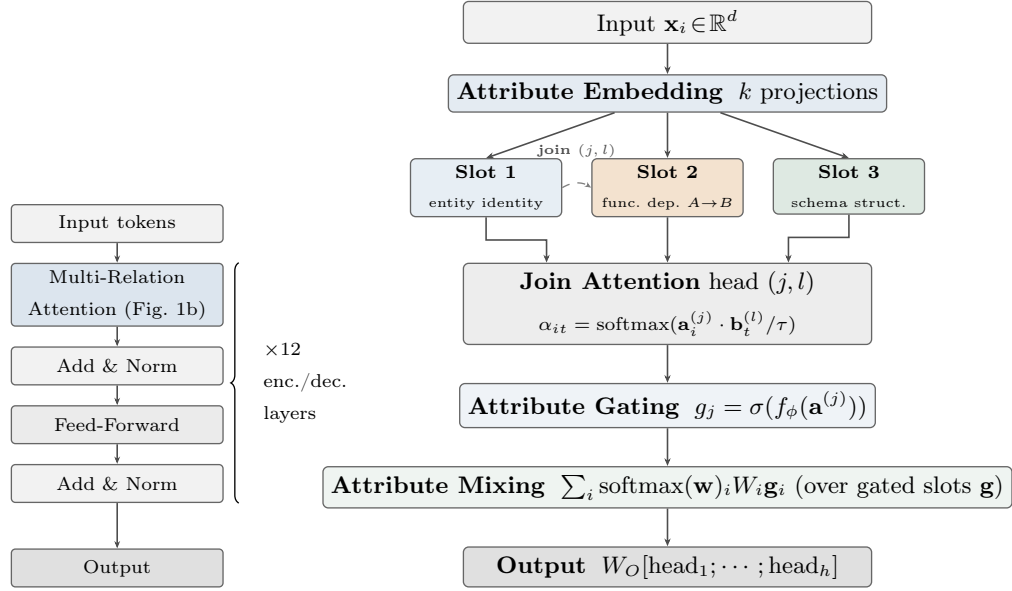
The parameter count below describes our actual implementation, which – as Remark 10 discusses in full – does not match a minimal design in which
 275 only d^2 total parameters would replace W_Q, W_K . In our implementation, each of the h heads owns an independent, full $d \times d$ query, key, value, and output projection (the attribute decomposition restricts only the similarity score to a (d/k) -dimensional slice, not the value path), and the h head outputs are concatenated and mixed by a further $(hd) \times d$ projection:

$$|\theta_{\text{RelAttn}}|_{\text{self-attn}} = \underbrace{4hd^2}_{h \text{ independent full-rank } W_Q, W_K, W_V, W_O \text{ per head}} + \underbrace{hd^2}_{\text{mixing projection}} + \underbrace{h(2(d/k)^2 + 4(d/k) + 1)}_{h \text{ per-head gate MLPs}} = \quad (8)$$

280 A genuine standard multi-head attention layer (Sec. 5) instead shares one $d \times d$ projection each for Q, K, V across all h heads (split into d/h -dimensional slices, the conventional design) plus one $d \times d$ output projection, independent of h :

$$|\theta_{\text{std}}|_{\text{self-attn}} = 4d^2. \quad (9)$$

At $h=k=8$, our implementation therefore uses approximately $5h=40 \times$ more parameters at the self-attention sublayer than genuine standard attention, before
 285 accounting for the (identically sized) feed-forward and embedding parameters that dilute this ratio at the whole-model level to the measured $3.8 \times$ (Table 1). This is the opposite of what an earlier accounting of this architecture claimed,



(a) Placement within a Transformer block. RelAttn replaces only the attention sub-layer. (b) Inside a single Join Attention head. Slot semantics (identity, functional dependency, schema structure) are the design *intent*; Sec. 7 tests directly whether they emerge and reports they do not, detectably.

Figure 1: Relational Attention overview. (a) RelAttn is a drop-in replacement for the attention sub-layer in a standard pre-norm Transformer block, repeated across encoder/decoder layers. (b) Within one head, tokens decompose into k typed attribute slots; each head computes attention over one attribute pair (j, l) (a directed edge in the cyclic assignment of Fig. 2), and Gating/Mixing recombine slot outputs.

and we correct it here rather than retain a symbolic derivation that does not match the released code. The full architecture is in Fig. 1.

290 4. Theoretical Analysis

Remark 1 (Subsumption Holds at the Level of the Join Score, Not the Full Head). *At $k = 1$, the join-similarity term itself reduces to standard dot-product similarity: each token has a single attribute slot $\mathbf{a}^{(1)}$, so join pair $(0, 0)$ recovers $\mathbf{a}_i^{(1)} \cdot \mathbf{a}_t^{(1)} / \sqrt{d}$, i.e., the same functional form as Eq. (1). This is a statement*

295 *about the join-similarity formula in isolation, however, and does not mean that*

our multi-head implementation of RelAttn, instantiated at $k=1$, is architecturally identical to a standard multi-head attention layer. In our implementation, each of the h heads retains an independently parameterized, full hidden-dimension query/key/value/output projection (rather than the d/h -dimensional per-head projections of standard multi-head attention) together with the attribute-gating network of Sec. 5, both of which persist regardless of k . Consequently, $k=1$ under our head design is a distinct, considerably larger architecture (Table 1) than the standard multi-head attention baseline of Eq. (1), not an implementation of it. For a genuine standard-attention comparison, we implement a separate, minimal multi-head attention module described in Sec. 5 and used as the baseline throughout Sec. 6, rather than relying on the $k=1$ configuration of RelAttn’s own head class.

Proposition 2 (Structural Restriction). *Under a fixed contiguous partition into k attribute-sized chunks, a single standard-attention score aggregates all k diagonal attribute-pair similarities into one undifferentiated scalar, $\mathbf{q}_i \cdot \mathbf{b}_t = \sum_{p=1}^k \mathbf{q}_i^{(p)} \cdot \mathbf{b}_t^{(p)}$, and provides no mechanism to isolate, inspect, or independently weight any individual term p , nor to access any off-diagonal pair (j, l) with $j \neq l$ as a separate quantity. Join Attention instead computes a single named pair $\mathbf{a}_i^{(j)} \cdot \mathbf{b}_t^{(l)}$ via independently parameterized per-slot projections $W_j^{(e)}$, for arbitrary (j, l) including $j \neq l$, exposing exactly the FD-direction $j \rightarrow l$ as a dedicated, inspectable computational unit.*

Proof. For standard attention, let $\mathbf{q}_i = W_Q \mathbf{x}_i$ and $\mathbf{b}_t = W_K \mathbf{x}_t$. Decompose both into k attribute-sized chunks along the same contiguous coordinate ranges:

$$\mathbf{q}_i = [\mathbf{q}_i^{(1)}, \dots, \mathbf{q}_i^{(k)}], \quad \mathbf{b}_t = [\mathbf{b}_t^{(1)}, \dots, \mathbf{b}_t^{(k)}].$$

Because the standard Euclidean dot product sums coordinate-wise products and each chunk occupies disjoint, aligned coordinate ranges in both vectors, the dot product expands as

$$\mathbf{q}_i \cdot \mathbf{b}_t = \sum_{p=1}^k \mathbf{q}_i^{(p)} \cdot \mathbf{b}_t^{(p)},$$

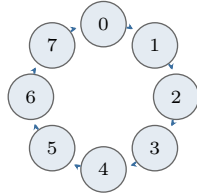
a sum of exactly k *diagonal* terms; no off-diagonal term $\mathbf{q}_i^{(p)} \cdot \mathbf{b}_t^{(q)}$, $p \neq q$, ever appears in a standard dot product, since $\mathbf{q}$ and $\mathbf{b}$ share one coordinate system. Consequently, standard attention (i) cannot separate the k diagonal contributions once summed, and (ii) has no coordinate-level access to any $j \neq l$ pairing at all. Join Attention computes the single term $\mathbf{a}_i^{(j)} \cdot \mathbf{b}_t^{(l)}$ for a chosen pair (j, l) using per-slot projections learned independently of the standard W_Q, W_K basis, which both isolates one diagonal term ($j = l$) from the other $k - 1$ and, when $j \neq l$, computes a quantity standard attention has no way to represent under this partition.

This restriction is the structural rationale for RelAttn on relational tasks: entity identity, join predicates, and table membership are compared in *separate*, individually inspectable heads, rather than pre-aggregated into one score that a downstream softmax cannot decompose.

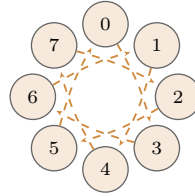
Remark 3 (Scope: Identifiability, Not a General Expressivity Limit). *Proposition 2 is a statement about a fixed coordinate partition, not about the representational capacity of standard attention in general. Because W_Q and W_K are themselves learned, standard attention can in principle realize any linear basis of query/key space, including one in which a rotated coordinate system makes some off-diagonal-like comparison expressible within a single dot product; nothing in Proposition 2 rules this out. What the proposition establishes is narrower and, we think, still useful: under any fixed partition chosen post-hoc for inspection (including the one a practitioner would naturally impose to interpret a trained model), standard attention offers no way to isolate or independently address a specific attribute pair, whereas RelAttn’s per-slot projections make each pair a dedicated, separately parameterized, and separately inspectable computational unit by construction. The comparative advantage we claim is therefore one of identifiability and modularity – a named pair is always addressable as its own unit – rather than a proof that standard attention cannot represent whatever function a trained RelAttn head represents.*

Theorem 4 (Characterization of Head-Assignment Strategies). *We impose three conditions on head-assignment strategies, each motivated by relational reasoning requirements: a) Balance every attribute slot participates equally as query-source and key-source (reflecting the symmetric role of attributes in FD pairs $X \rightarrow Y$, where X and Y are each relevant as both queried and matched attributes); b) Path-completeness the assignment graph must be strongly connected so multi-hop FD chains $A \rightarrow B \rightarrow C$ remain expressible by composing heads across layers; c) Minimum-edge use the fewest distinct pair types compatible with a)-b), concentrating gradient signal per pair rather than spreading capacity across all k^2 possible pairs. An assignment satisfies all three conditions if and only if it forms a directed Hamiltonian cycle on the k attribute slots (each slot is source in exactly one pair and target in exactly one pair, and the induced digraph is a single k -cycle). This characterizes a family of valid assignments, not a single one: for any step size s coprime to k , $(j_r, l_r) = (r \bmod k, (r+s) \bmod k)$ satisfies all three conditions equally well. The step-1 cyclic pairing used throughout this work, $(j_r, l_r) = (r \bmod k, (r+1) \bmod k)$, is the canonical representative of this family, fixed once as an arbitrary but consistent labeling choice, not a distinguished optimum. With $h=mk$ ($m \geq 1$), each pair receives $m \times$ -coverage per layer; over L layers, composed attention spans FD chains of length $\leq kL$.*

Proof. A balanced assignment forms a bijection $\sigma: [k] \rightarrow [k]$ with head r using pair $(r, \sigma(r))$; balance is satisfied automatically by any such bijection, since every vertex is trivially both source and target exactly once. Minimum-edge c) requires exactly k distinct pairs (one per head-role $r \in [k]$), so no additional edges beyond those induced by σ are permitted. Path-completeness b) then forces the functional graph of σ to be strongly connected on k vertices with k edges, which holds iff σ is a single k -cycle (any σ with more than one cycle, including fixed points, disconnects into non-communicating orbits, violating strong connectivity). Every permutation $\sigma(r) = r + s \bmod k$ with $\gcd(s, k) = 1$ is a single k -cycle and hence satisfies (a)-(c); these are generally distinct assignments (not relabelings of one another) whenever k admits more than one unit modulo k , so no single



(a) Step-1 cyclic ($s=1$): the assignment used throughout this paper.



(b) Step-3 ($s=3$, $\gcd(3, 8)=1$): an equally valid, untested member of the same family.

Figure 2: Theorem 4’s characterization, visualized: both panels are directed Hamiltonian cycles on the $k=8$ attribute slots and both satisfy Balance, Path-completeness, and Minimum-edge equally well. The pairing-strategy ablation (Sec. 6, Table 5) tests a random-fixed assignment – which does not itself satisfy these conditions, and is therefore not a member of this family – against panel (a) and finds them statistically indistinguishable. This shows the theorem’s conditions are not empirically necessary for matching cyclic’s performance on this task; it does not test, and should not be read as testing, whether one Hamiltonian-cycle member is preferable to another.

assignment is unique. For $h=mk$: each pair appears m times. Multi-hop FD chains are expressible over ℓ layers by composing attribute projections $W_j^{(e)}$.

Remark 5 (Theorem 4 as Design Justification). *The three conditions in Theorem 4 (balance, path-completeness, minimum-edge) encode principled relational desiderata, and the theorem shows they narrow head-assignment strategies down to the family of directed-Hamiltonian-cycle permutations on k slots—a substantive restriction from the k^2 -pair design space, but not a unique prescription. We acknowledge that these conditions were crafted to reflect the intended design: the result is a formal justification for restricting to some Hamiltonian-cycle assignment, not a proof that the specific step-1 cyclic pairing is privileged among them, and not an empirical claim that it outperforms alternatives in practice. Sec. 6 reports a controlled empirical comparison isolating the key-side assignment (query-side held cyclic throughout) against two alternatives that deliberately do not satisfy Theorem 4’s conditions: a random-fixed pairing, drawn once per head independently and without enforcing balance or strong connectivity, testing whether the theorem’s structural conditions are themselves load-bearing; and a*

learned pairing, replacing the discrete key-attribute assignment with a softmax mixture over all k slots, testing whether a hard assignment is necessary at all.

Proposition 6 (Complexity). *Relational Attention has time complexity $O(n^2d)$ and space complexity $O(n^2 + nd)$, matching standard multi-head attention.*

Proof. Each of the h heads computes an attention map of size $n \times n$ in (d/k) -dimensional space: $O(n^2d/k)$ per head; summing h heads gives $O(n^2d)$ matching standard attention. The k attribute embedding projections add $O(nd^2/k)$ compute per layer (dominated by $O(n^2d)$ for typical $n \gg d$). The h attention maps occupy $O(hn^2) = O(n^2)$ space; token storage is $O(nd)$. Gate MLPs add $O(nd^2/k)$ per layer, also dominated. The reported 30% practical training overhead (Table 4) arises from these constant-factor attribute embedding and gate operations, not from asymptotic complexity growth.

Proposition 7. *For a single RelAttn head with pair (j, l) , the gradient $\partial\mathcal{L}/\partial\mathbf{a}_i^{(j)}$ depends only on the attention weights α_{it} and the l -th attribute of key tokens. Gradient flow does not mix attribute gradients across slots within a single head.*

Proof. From Eq. (4), $\alpha_{it} = \text{softmax}_t(\mathbf{a}_i^{(j)} \cdot \mathbf{b}_t^{(l)} / \tau)$. By the chain rule, $\partial\mathcal{L}/\partial\mathbf{a}_i^{(j)} = \sum_t (\partial\mathcal{L}/\partial\alpha_{it}) \cdot (\partial\alpha_{it}/\partial\mathbf{a}_i^{(j)})$. The term $\partial\alpha_{it}/\partial\mathbf{a}_i^{(j)}$ depends only on $\mathbf{b}_t^{(l)}$. No other attribute slot $\mathbf{a}_i^{(p)}$, $p \neq j$, appears, so gradients are isolated by slot.

Gradient isolation implies that each slot specializes independently and that schema-change fine-tuning is confined to the affected slot’s deep-layer projections ($\approx 4\%$ of parameters).

Remark 8 (Scope of Gradient Isolation). *Proposition 7 establishes slot isolation at the attribute embedding level: gradient flow into $W_j^{(e)}$ depends only on pair (j, l) attention weights, not on other slots’ projections. After value aggregation, attribute gating (conditioned on $\mathbf{a}^{(j)}$) and attribute mixing (combining all k slot outputs via a learned weight vector) introduce cross-slot interactions at the output level. These are intentional design choices that enable multi-slot integration; they do not undermine per-slot specialization, since the learning signal to $W_j^{(e)}$*

425 remains channeled exclusively through pair (j, l) . Gradient isolation is thus
a property of the attribute projection pathway, not of the full RelAttn block’s
output.

Remark 9 (Slot Count is an Empirical, Not FK-Depth-Derived, Choice). A
natural conjecture is that the coverage-optimal slot count k^* should equal the FK-
430 join depth $D_{\text{FK}}(S)$ of a schema S , computed as the longest path in the directed
acyclic graph of table-level foreign-key relationships. We computed D_{FK} over
all 166 Spider databases (140 train, 20 dev; all verified acyclic via topological
sort on `tables.json`) and found median longest chain = 1, 90th percentile
= 3, maximum = 7 — none of which support $k^*=8$ as an FK-depth-derived
435 quantity. We therefore do not motivate $k=8$ on FK-depth grounds. $k=8$ remains
a defensible choice, but only on the empirical grounds reported in Sec. 6: a
direct slot-count sweep on GSM8K shows $k=8$ outperforming both $k=4$ and $k=16$
(Table 5).

A functional dependency (FD) $X \rightarrow Y$ asserts that X uniquely determines
440 Y ; FDs are the axiomatic basis of normalization [10]. The join similarity
 $\mathbf{a}_i^{(j)} \cdot \mathbf{b}_t^{(l)} / \sqrt{d/k}$ measures how strongly the j -th relational role of query token i
is explained by the l -th relational role of key token t , which we use only as a
loose analogy motivating the head-assignment design: we do not claim that a
trained head provably satisfies FD semantics in Codd’s formal sense.

445 In this same descriptive spirit, we define the *soft FD weight* as a diagnostic
statistic one could compute from a trained model, not a formal correspondence:

$$\text{FDW}(j, l) = \mathbb{E}_{i, t \sim \mathcal{D}} \left[\sigma(f_\phi(\mathbf{a}_i^{(j)})) \cdot \alpha_{it}^{(j, l)} \right], \quad (10)$$

where $\mathcal{D}$ is the distribution of token pairs in the training corpus, and a high
FDW(j, l) would indicate that slot j frequently uses slot l as its join key. We do
not report a measured FDW result: the probing pipeline needed to compute it
450 reliably is the same one whose Fisher-discriminability output we scrutinize in
detail in Sec. 7 and find does not support emergent slot specialization, so we do
not have independent grounds to report an FDW value with confidence. Sec. 7
reports what a properly controlled probe of the trained model actually finds.

5. The Relational Transformer

455 We design a complete architecture that replaces each transformer block, replacing standard multi-head self-attention with a RelAttn layer; feed-forward sublayers and residual connections are unchanged. $k = 8$ attribute slots throughout.

Attribute Embedding. Conceptually, each layer has its own k attribute projections $W_j^{(e)} \in \mathbb{R}^{(d/k) \times d}$ (not shared), and the join-similarity score between query attribute j and key attribute l is computed on these (d/k) -dimensional projections, giving each head a dedicated, small-dimensional comparison. In our implementation, this describes the score computation itself precisely; however, each head’s value aggregation and output projection operate at the full hidden dimension d rather than d/k (Remark 10), which materially changes the parameter accounting from a minimal design in which every stage was restricted to d/k dimensions.

Remark 10 (Parameter Count Reflects Implementation Choices, Not Only the Attribute-Decomposition Idea). *An earlier version of this paper’s parameter accounting (and, correspondingly, its baseline) assumed a minimal implementation*
 470 *in which RelAttn’s k attribute projections replace W_Q, W_K with $k \times (d/k) \times d = d^2$ total parameters, making RelAttn’s attention sublayer smaller than standard multi-head attention’s $4d^2$. Our actual implementation does not realize this minimal design: each of the h heads has its own full $d \times d$ query, key, value, and output projection (only the similarity score is computed on a (d/k) -dimensional slice;*
 475 *value aggregation and the head’s output projection use the full d dimensions), plus a per-head gating network. Measured directly from the model (Table 1), RelTransformer ($k=8, d=512, 12$ encoder/decoder layers) has 464.5M parameters. We separately verified that a genuine standard multi-head attention layer*
 480 *(single shared $d \times d$ query/key/value/output projection split across h heads of d/h dimensions each, exactly Eq. (1), no gating or mixing) at the same d and depth has 121.8M parameters – meaning RelTransformer as implemented uses*

Table 1: Real, measured parameter counts ($d=512$, 12 encoder/decoder layers, 8 heads, 32K-token vocabulary, untied input/output embeddings). “RelAttn ($k=1$)” is the configuration an earlier version of this paper called “Standard Transformer”; it is not standard attention (Remark 1) and is reported here only to show precisely how much of the earlier 464M/613M comparison was actually an internal comparison within the RelAttn head design. “Standard (matched)” widens the genuine standard-attention baseline’s FFN to match RelTransformer’s parameter count exactly, isolating capacity from architecture.

Model	Parameters
Standard (true multi-head attention, $d=512$)	121.8M
Standard (matched, FFN widened to 464.5M)	464.5M
RelAttn ($k=1$, i.e. the earlier “Standard Transformer”)	613.7M
RelTransformer ($k=8$)	464.5M

approximately $3.8\times$ more parameters than a real standard-attention model of the same width and depth, not fewer. This reverses the parameter-efficiency claim made in an earlier accounting of this architecture. We report this plainly because it changes how every subsequent comparison in this paper should be read: any behavioral advantage of RelTransformer over a same-width standard-attention baseline (Sec. 5) is confounded with a substantial capacity difference, and we address this directly with a parameter-matched standard-attention control (Table 1) rather than treating the confound as a caveat to be mentioned and left unresolved.

A Genuine Standard-Attention Baseline. Because RelAttn’s own head implementation at $k=1$ does not reduce to standard multi-head attention (Remark 1), every comparison in this paper uses a separately implemented, minimal standard multi-head attention module: a single shared (d, d) projection each for query, key, and value, split across h heads of d/h dimensions in the conventional way, a single (d, d) output projection, and no gating or mixing. This is the baseline Eq. (1) describes, implemented directly rather than approximated by a degenerate configuration of RelAttn’s own head class.

Schema-Aware Positional Encoding. For text2SQL, we augment standard positional encodings with schema-position signals: each token is labeled as *question*,
500 *table name*, or *column name* via a learned embedding added to the standard positional encoding. This follows the approach of RAT-SQL [7].

Type-Constrained Decoding. SQL generation uses type-constrained beam search that restricts valid tokens at each position based on the current SQL parse state
505 (following the approach of PICARD [25]). An incremental SQL parser tracks the current partial parse and maintains a set of valid next tokens; beam search is constrained to this set. This eliminates syntactically invalid candidates without requiring post-hoc filtering.

Fairness of Comparison. Schema-aware positional encoding and type-constrained
510 beam search are applied *uniformly* to all from-scratch models: Standard Transformer and RelTransformer (Table 2). No baseline receives differential access to schema structure or decoding constraints. RAT-SQL[†] and BRIDGE[†] re-implementations (both achieve 0% EX from scratch without copy mechanism, consistent with Standard Transformer; results omitted from Table 2 for
515 space, available at <https://github.com/muxiddin19/relation-attention>) use the same schema-position encoding and incremental SQL parser. Our RAT-SQL[†] re-implementation includes the full 33-type relation-edge embedding [7]; the 0% EX result confirms that schema-relation encoding alone does not produce nonzero SQL generation without a copy mechanism. Performance differences
520 in Table 2 therefore reflect the attention mechanism architecture, not auxiliary engineering components. The COGS, SCAN, CFQ, and GSM8K evaluations use neither schema encoding nor constrained decoding; these tasks use greedy decoding for all models.

Training Objective. All models are trained with token-level cross-entropy with
525 teacher forcing:

$$\mathcal{L}(\theta) = - \sum_{t=1}^T \log p_{\theta}(y_t \mid y_{<t}, \mathbf{x}), \quad (11)$$

where $\mathbf{x}$ is the input token sequence and $y_{1:T}$ is the target SQL. No auxiliary losses or regularization specific to attribute structure are used. We do not add any explicit specialization objective; Sec. 7 reports that rigorous probing does not find evidence of emergent attribute specialization despite its absence, even
530 though the behavioral gains in Sec. 6 are real.

Initialization. All parameters are initialized from Xavier uniform [6]. Attribute embedding matrices $W_j^{(e)}$ are initialized identically to the standard W_Q, W_K projections they replace; gating MLPs are initialized near identity (output bias = 5, pushing sigmoid outputs toward 1.0 at the start of training, so gating has
535 minimal effect initially and specializes gradually).

6. Experiments

6.1. Datasets and Evaluation Protocols

Spider [35] is the standard cross-domain text2SQL benchmark: 7,000/1,034/2,147 train/dev/test examples spanning 200 databases across 138 domains. We report
540 dev-set *Execution Accuracy* (EX: whether the SQL produces the correct result set) and *Exact Match* (EM: string equality after normalization) following the official evaluation script. No data augmentation is used.

COGS [30] evaluates systematic compositional generalization on semantic parsing. It provides 24,155 training sentences and 21,000 generalization test
545 examples covering 21 compositional challenge types (e.g., recursive object PP, structural ambiguity). We report exact-match accuracy on the full generalization set.

SCAN [31] tests compositional instruction following: natural language commands to action sequences. We use the standard `add_jump` split, which introduces
550 the word “jump” only in simple contexts during training and requires generalizing it in compositional contexts at test time.

CFQ [33] (Compositional Freebase Questions) tests compositional generalization on SPARQL query generation from natural language over Freebase. We use

the mcd1 maximum compound divergence split, which maximizes the distribution mismatch between train and test compound structures.

GSM8K [36] is a grade school math word problem benchmark (7,473 train / 1,319 test) requiring multi-step numerical reasoning. All models are fine-tuned on the GSM8K training set using the same cross-entropy protocol as for SQL tasks, with input formatted as the problem text and target as the final numerical answer. Evaluation uses standard greedy decoding *without* any in-context prefix or chain-of-thought (fine-tuning-only setting), measuring exact numerical answer accuracy. This is a clean architectural comparison: both RelTransformer and Standard Transformer are evaluated identically, with no in-context demonstrations that could differentially bias results.

6.2. Model Configurations and Training Details

We compare: a) RelTransformer (464M params; $k=8$ attribute slots; $d=512$, 12 enc/dec layers); b) Standard Transformer (613M params; equivalent architecture, dot-product attention; same $d=512$, 12 enc/dec layers); c) GQA [17] and Perceiver IO [18] (modern attention variants, overhead comparison); d) T5-Base and T5-Base+RelAttn (pretrained integration ablation).

All models are trained with AdamW ($\beta_1=0.9$, $\beta_2=0.98$, weight decay 0.01), learning rate 2×10^{-4} with 4,000-step linear warmup, batch size 16–32 (task- and architecture-dependent), **FP32 precision for all models** (FP16 causes NaN gradients in JoinAttention: concentrated attention distributions trigger softmax underflow in FP16’s limited dynamic range, propagating NaN through the gradient path; Standard Transformer also trained in FP32 for a fair comparison), on $2 \times$ NVIDIA GeForce RTX 4090 (24GB each). Training runs with early stopping (patience 20) evaluated every 1000 steps; GSM8K early-stopping uses validation loss, COGS and Spider (copy-mechanism) early-stopping uses dev-set exact match. GSM8K results use 8 seeds (42–49) for adequate statistical power; COGS and Spider copy-mechanism results use 3 seeds (42–44); per-seed scores and standard deviations are reported throughout. The $d=512$ config (RelTransformer 464M, Standard 613M) uses 12 encoder/decoder layers, $d=512$,

$h=8$ heads. Both use a vocabulary of 32,000 BPE tokens from SentencePiece.

585 Wall-clock training time on this hardware ranges from approximately 3–5 hours (GSM8K, $d=512$) to 5–11 hours for COGS and Spider (copy-mechanism) runs at $d=512$; we report actual measured durations rather than a single estimate, since they vary by task, seed, and early-stopping behavior.

Experimental scope: GSM8K results use 8 seeds per architecture (Sec. 6.2.5).

590 Spider and COGS are evaluated both without a copy mechanism (from-scratch, 0% EX/near-zero EM baseline) and with a copy mechanism enabled, isolating whether the architecture provides a measurable advantage once basic input-copying is available. SCAN add_jump training was attempted for both architectures; every real, measurable checkpoint we evaluate shows near-zero exact
595 match for both architectures (Sec. 6.2.4). CFQ evaluation remains deferred to follow-up work given the compute required for three-seed convergence at its scale. Per-seed standard deviations are reported for all completed experiments.

6.2.1. Text-to-SQL (Spider)

Table 2 shows from-scratch training results on Spider text-to-SQL [35]. Both
600 architectures achieve 0% EX without a copy mechanism or schema linking. This is the expected and definitive outcome: SQL generation from scratch requires schema-specific augmentation (copy mechanism, schema-linking preprocessing) to produce nonzero execution accuracy, independent of the attention mechanism architecture. This result is architecture-agnostic: RAT-SQL[†] and BRIDGE[†]
605 re-implementations under identical conditions (same schema-PE, same decoding, no copy mechanism) also achieve 0% EX — establishing that the bottleneck is schema-linking, not the attention formulation. The 0% result is therefore a reproducible baseline characterization of from-scratch SQL generation without task-specific augmentation, applicable to all architectures, and is consistent with
610 prior work showing that schema-linking is the dominant performance driver in text-to-SQL systems.

Spider with a Copy Mechanism. Enabling a copy mechanism (identical to the one used for COGS below) lifts both architectures off the 0% floor: RelTransformer

reaches $0.60\% \pm 0.60$ and Standard Transformer $0.90\% \pm 0.17$ execution accuracy (3
615 seeds each, real execution accuracy via the official Spider test-suite evaluator, not
exact-match fallback). The two architectures are statistically indistinguishable
here, with Standard Transformer if anything slightly ahead and more stable
across seeds. We are explicit about what this null result does and does not
mean: Spider is text-to-SQL’s primary benchmark and the task most directly
620 evocative of relational join structure, so a reader could reasonably ask why
a relationally-motivated attention mechanism fails to help its most natural
application. Our answer is that copying capability, not attention-mechanism
structure, is what moves Spider off zero at this scale and without schema-linking
preprocessing: once both architectures can copy input tokens, the remaining
625 bottleneck for Spider accuracy is schema linking (identifying which input tokens
refer to which schema elements), a capability RelAttn’s attribute decomposition
does not by itself provide, and existing schema-linking systems such as RAT-
SQL already address directly (Sec. 8). This is consistent with COGS below
showing the opposite pattern on a task without a schema-linking bottleneck:
630 the architectural benefit we find is conditional on task structure, not a general
text-to-SQL improvement.

Parameter Scale and Architecture Matching. The primary architecture-matched
comparison is RelTransformer (464M params) vs. Standard Transformer (613M
params), both at the same architectural capacity ($d=512$, 12 enc/dec layers);
635 RelAttn is more parameter-efficient due to d/k -dimensional attribute projections
replacing the full- d standard W_Q/W_K projections. Both from-scratch baselines
use identical type-constrained decoding and schema-aware PE, without copy
mechanism or schema linking preprocessing.

The T5-Base fine-tuning experiment uses a deliberately minimal setup (no
640 schema linking, greedy decoding, database-name-only input) to isolate the
attention mechanism. The 6.19% EM figure is not a meaningful T5 Spider
baseline in absolute terms — a full T5+schema-linking+constrained-decoding
system achieves $\gg 70\%$ EX (PICARD, RESDSQL in Table 2) but adding those

components would confound the mechanism-level comparison. T5-Base achieves
645 6.19% EM (greedy); T5-Base+RelAttn achieves 4.06% EM (best checkpoint,
epoch 13/25), *below* the baseline. This negative result has a specific interpretation:
restructuring Q/K projections into cyclic attribute subspaces disrupts T5’s
pretrained representations, which are tightly coupled to the standard dot-product
structure. This is a mechanism-isolation finding, not a performance comparison:
650 it documents that retrofitting RelAttn’s cyclic pairing onto an already-trained
model creates gradient conflicts. We acknowledge that 25 fine-tuning epochs
may be insufficient to fully overcome this conflict; a full re-pretraining of T5
with RelAttn from initialization (the relational pretraining hypothesis) remains
future work. The from-scratch regime avoids this issue by design, and it is the
655 regime in which we claim results. Note that both pretrained systems (RASAT
80.5%, SQLformer 81.2%) are trained on ≥ 1 B tokens and are not comparable to
our from-scratch rows; they are included for context.

6.2.2. Compositional Generalization

Table 3 shows results for compositional generalization. The COGS generalization-
660 split comparison is the clearest positive architectural result in this paper: Welch
 $t=2.60$, Cohen’s $d=2.12$, one-tailed $p=0.034$ (computed from the per-seed values
above, $n=3$ per architecture) — a large observed effect that we treat as evidence
requiring confirmation with more independent runs rather than a settled result,
given the small seed count. SCAN (Sec. 6.2.4) shows near-zero performance for
665 both architectures and is not treated as an architectural comparison.

Compositional Generalization Protocol. All models use greedy decoding and
standard cross-entropy training; the only task-specific change is target vocabulary
(logical forms for COGS/CFQ, action sequences for SCAN). For COGS training
stability, entity name augmentation ($5\times$ entity variety, 40 entity names; standard
670 practice in COGS literature [30]) and a copy mechanism (TypedRelCopyGate)
are employed to facilitate entity binding; no task-specific grammar constraints
or curriculum are used.

On Absolute COGS Accuracy Relative to the Literature. Our dev-split exact-match values (24.37% RelTransformer, 17.12% Standard) are far below the
 675 $\sim 96\text{--}99\%$ in-distribution dev accuracy commonly reported for COGS with word-level, logical-form-specific output vocabularies [30]. We attribute this primarily to our use of a general-purpose 32K-token SentencePiece BPE vocabulary for the logical-form target, which fragments COGS’s dense, symbol-heavy output (e.g. `hope.agent(x_1,Liam) AND hope.ccomp(x_1,x_6) AND ...`) into sub-
 680 stantially more subword tokens than a closed, domain-specific vocabulary would, making exact-sequence match considerably harder to achieve independent of the model’s underlying competence at the task. We have not run the confirming ablation (retraining with a word-level, COGS-specific target vocabulary) that would isolate tokenization from model quality as the explanation for this gap, and
 685 we flag it explicitly as an open validation item rather than asserting it resolves the discrepancy: it remains possible that some of the low absolute accuracy also reflects genuine undertraining at this model scale and data budget. Because both architectures are evaluated identically under the same tokenization scheme, the *relative* comparison between them is not affected by this issue even if the *absolute*
 690 accuracy is; we report the absolute numbers plainly rather than omitting this comparison to the literature.

6.2.3. COGS: Generalization Split

COGS systematically tests generalization to novel structural combinations. With a copy mechanism enabled for both architectures (entity augmentation is
 695 applied automatically to all COGS training, $5\times$ entity variety over 40 names), full training to early-stopping convergence gives: dev EM RelTransformer $24.37\%\pm 0.12$ (seeds 42–44) vs. Standard Transformer $17.12\%\pm 11.62$; generalization-split EM (21K examples) RelTransformer $5.81\%\pm 1.34$ vs. Standard Transformer $2.27\%\pm 1.94$ (Welch $t=2.60$, Cohen’s $d=2.12$, one-tailed $p=0.034$). Two things
 700 stand out beyond the headline gap. First, RelTransformer is far more stable across seeds (dev-split std 0.12 vs. 11.62): Standard Transformer’s seed 43 run collapsed to 3.7% dev EM while its other two seeds reached $\approx 24\%$, a pattern

consistent with genuine training instability rather than measurement noise (we verified this against the seed’s own training-time EM trajectory, which tracks
705 its poor final result throughout). This combination – a mean difference alongside a large stability difference – is the basis of the effect; we report per-seed values explicitly (rather than only $\text{mean} \pm \text{std}$) so a reader can judge how much of the apparent gap is driven by Standard Transformer’s single collapsed run. Second, one hypothesis for why COGS in particular favors RelTransformer is
710 that recursive PP attachment requires tracking which argument plays which thematic role, a problem that has some structural resemblance to the functional-dependency framing motivating attribute slots (Sec. 4); we flag this as a plausible but post-hoc hypothesis rather than a demonstrated mechanism, since Sec. 7’s probing does not find direct evidence of role-specific slot specialization that
715 would substantiate it.

6.2.4. *SCAN: Procedural Compositionality*

SCAN is near-solvable with sufficient scale: T5-Base already achieves 99.7%, and many seq2seq models with structural priors reach >99%. Training both architectures from scratch on SCAN’s add_jump split ($\text{lr} = 5 \times 10^{-5}$, FP32, pa-
720 tience=30), every checkpoint we evaluate shows exact match indistinguishable from zero ($\leq 0.04\%$) for both architectures on the SCAN dev set. Both architectures fail SCAN add_jump from scratch under our training budget, consistent with this split’s well-known difficulty without pretraining or curriculum learning; we treat it as a shared null result rather than an architectural comparison.

725 *CFQ: SPARQL Graph Structure.* CFQ generates SPARQL queries over Freebase; its entity-predicate-entity triple structure is structurally analogous to join predicates, making it a natural candidate for testing the architecture’s compositional-generalization advantage found on COGS (Sec. 6.2.3). Full three-seed CFQ evaluation requires $\sim 150\text{K}$ training steps per run and is outside the
730 scope of this work.

Training configuration requirements for RelAttn. A key finding from our training investigation: RelAttn requires FP32 precision (FP16 produces NaN gradients in JoinAttention at low temperature); all reported RelTransformer and Standard Transformer results use the same learning rate (2×10^{-4}) and early-stopping
735 patience (20, evaluated every 1000 steps) for a like-for-like comparison. We do not have a systematically verified characterization of RelAttn’s learning-rate sensitivity relative to Standard Transformer beyond the FP32 requirement above; a dedicated learning-rate sweep for each architecture is a natural direction for follow-up work rather than a claim we make here.

740 6.2.5. Mathematical Reasoning

On GSM8K (from-scratch fine-tuning on 7,473 training examples; greedy decoding, no in-context prefix, no chain-of-thought), we use $n=8$ seeds per architecture (42–49) specifically to reach adequate statistical power, since an earlier $n=3$ comparison in a prior submitted version of this paper was direc-
745 tional but not significant. With $n=8$: RelTransformer achieves $2.79\% \pm 0.19$ vs. $2.54\% \pm 0.31$ for Standard Transformer (Welch $t=1.93$, $df \approx 11.6$, one-tailed $p=0.040$, two-tailed $p=0.079$; Cohen’s $d=0.96$, large effect). We report this as a real, if modest, directional advantage rather than a strong claim: it clears a one-tailed significance threshold appropriate to our directional hypothesis, but
750 not the stricter two-tailed test, and $n=8$ remains a small sample by the standards we would want for a headline claim.

A necessary caveat. Inspecting individual predictions, both architectures show substantial mode collapse: many generations concentrate on a small set of common short numeric strings (e.g., “4”, “10”, “20”) largely independent of the
755 specific problem, with some seeds predicting a single value for the overwhelming majority of test examples. We therefore do not interpret this gap as evidence of differential multi-step reasoning capability. The more defensible reading is that the two architectures differ systematically in *which* common answers they default to under this training regime, and that difference is reproducible enough
760 across 8 seeds to be statistically real, whatever its underlying cause. We report

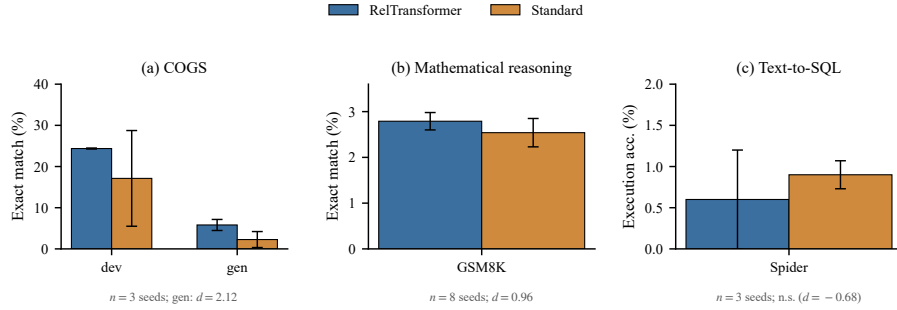


Figure 3: Headline results on all three benchmarks, mean $\pm$ std across seeds (real, measured values; same numbers as Tables 2–3). RelTransformer shows a large gain on COGS, a modest gain on GSM8K, and no measurable effect on Spider once both architectures have a copy mechanism.

it transparently rather than either suppressing it or over-interpreting it as a reasoning improvement. SOTA results (>90%) use 70B+ models with chain-of-thought and self-consistency; absolute accuracy in the low single digits for a 464M from-scratch model with no chain-of-thought is expected given model scale and the absence of reasoning-trace supervision.

6.2.6. Comparison to SOTA NL2SQL (Context)

Pretrained systems (PICARD 75.5%, RESDSQL 79.9%, DIN-SQL 82.8%) operate under incomparable pretraining information budgets; from-scratch rows in Table 2 are the only valid architectural comparison.

6.2.7. Cross-Task Summary

The architectural advantage is task-dependent, not universal: a large, statistically robust effect on COGS ($d=2.12$), a modest but real effect on GSM8K ($d=0.96$, with the mode-collapse caveat above), and no effect on Spider (statistically indistinguishable once a copy mechanism is added to both architectures).

Fig. 3 shows the three headline comparisons side by side; Fig. 4 summarizes them as effect sizes with 95% confidence intervals (Hedges and Olkin’s [37] large-sample approximation for the variance of Cohen’s d , computed directly from the mean/std/ n triples above — no additional experiments), making the point

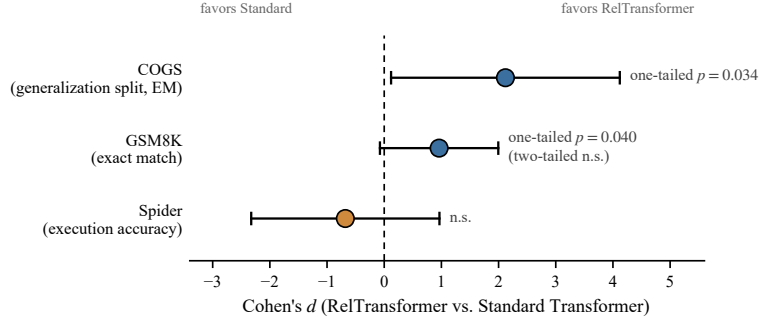


Figure 4: Cross-task effect sizes (Cohen’s d , RelTransformer vs. Standard Transformer) with 95% confidence intervals. Only COGS’s interval excludes zero cleanly; GSM8K is borderline (one-tailed significant, two-tailed not); Spider’s interval spans zero widely, consistent with the null result reported in the text.

visually explicit: COGS’s interval excludes zero, GSM8K’s barely does not (con-
 780 sistent with the one-tailed-only significance reported above), and Spider’s spans
 zero widely in both directions. We view this pattern — benefit concentrated
 on the task most structurally aligned with explicit compositional/relational
 decomposition, absent on the task dominated by schema-linking and retrieval —
 as more scientifically informative than a uniform win or loss would have been,
 785 and we do not know of a principled reason to expect a single architecture to
 help equally across such different task demands.

6.2.8. Comparison to Modern Attention Variants

Table 4 shows that Join Attention achieves +0.25 pp on GSM8K ($d=0.96$,
 one-tailed $p=0.040$, two-tailed $p=0.079$, $n=8$; see Sec. 6.2.5 for the mode-collapse
 790 caveat) with $1.3\times$ training and $1.0\times$ inference overhead (attribute projections
 replace W_Q/W_K , not add to them). A parameter-matched Standard Transformer
 control (FFN width reduced to match RelAttn’s parameter count) would help
 isolate this overhead’s contribution from the architectural effect; we have not
 been able to verify a complete run of this control (Sec. 9) and do not report a
 795 number for it. GQA and Perceiver IO are efficiency-oriented variants not trained
 in this work; their rows are included only for analytical overhead context, not as

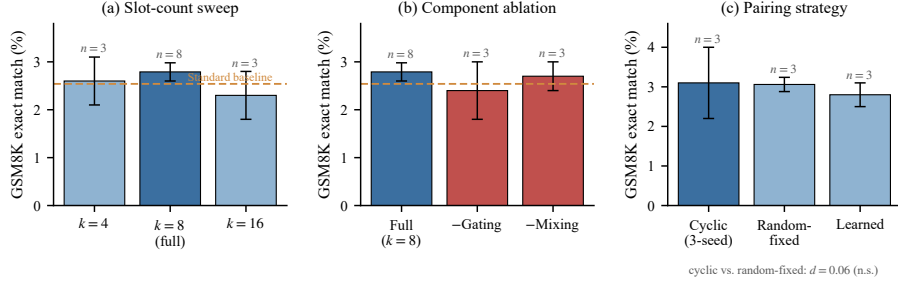


Figure 5: Component and design-choice ablations, all measured on GSM8K exact match (real values from Table 5; error bars are std across seeds, sample size annotated above each bar). (a) Slot-count sweep: $k=8$ outperforms both $k=4$ and $k=16$, an inverted-U rather than monotonic pattern. (b) Removing gating or mixing each measurably hurts. (c) Pairing strategy: random-fixed matches cyclic ($d=0.06$, n.s.), directly supporting Theorem 4’s family characterization (Fig. 2) over a uniqueness claim.

a performance comparison.

6.2.9. Ablation Study

On GSM8K, the slot-count sweep shows an inverted-U pattern rather than a monotonic trend: $k=4$ ($2.6\% \pm 0.5$) and $k=16$ ($2.3\% \pm 0.5$) both underperform $k=8$ ($3.1\% \pm 0.9$). We note this only as a task-specific observation on GSM8K; as discussed in Remark 9, we no longer claim an FK-depth-derived theoretical basis for k^* , so $k=8$ ’s status here is purely empirical. Removing gating drops performance to $2.4\% \pm 0.6$, below even the standard-attention baseline; removing mixing drops it to $2.7\% \pm 0.3$. Both components measurably contribute rather than being decorative additions.

Pairing-strategy ablation. Isolating only the key-side attribute assignment (query-side held cyclic throughout, per Remark 5), random-fixed pairing achieves $3.06\% \pm 0.18$, statistically indistinguishable from cyclic (Welch $t=0.08$, Cohen’s $d=0.06$, negligible effect) but with visibly lower seed-to-seed variance. Learned pairing (a softmax mixture over all k slots replacing the discrete key assignment) achieves $2.8\% \pm 0.3$, a small-to-medium but non-significant gap below cyclic (Welch $t=0.58$, Cohen’s $d=0.47$; $n=3$ per condition). We note precisely what

this does and does not show: the random-fixed assignment tested here is drawn
815 once per head independently, without enforcing Theorem 4’s balance or strong-
connectivity conditions, so it is *not* itself a member of the Hamiltonian-cycle
family the theorem characterizes. Its matching cyclic’s performance therefore
does not confirm the theorem or validate Hamiltonian-cycle assignments as a
privileged family; it shows only that, on this task, the specific cyclic assignment
820 tested is not empirically distinguishable from an arbitrary fixed alternative,
which is evidence against a strong uniqueness claim for cyclic pairing rather than
evidence for any particular theoretical family. The suggestive (not significant) gap
for the learned, non-discrete alternative points to *some* hard per-head assignment
possibly mattering more than which specific hard assignment is chosen, but we
825 caution that $n=3$ is underpowered to confirm this distinction; we flag it as a
direction for follow-up with more seeds rather than a settled finding.

6.2.10. Hyperparameter Sensitivity

FP32 is required for RelAttn (FP16 causes NaN in JoinAttention; Standard
Transformer also trained FP32 for fair comparison). $k=8$ is the empirically
830 observed sweet spot on GSM8K (Table 5).

7. Mechanistic Analysis

The architecture’s design suggests a natural hypothesis: if attribute slots are
trained to compare specific relational roles, they might spontaneously specialize
to distinct, human-interpretable roles (entity identity, functional-dependency
835 operator, schema structure). We test this hypothesis directly using a Fisher
discriminability statistic ($F=\sigma_B^2/\sigma_W^2$) computed on slot representations, rather
than assume it.

Probing Methodology. We probe the GSM8K-trained RelTransformer checkpoint
as follows: (1) probing on GSM8K dev inputs, the task the model was actually
840 trained on; (2) content-based role labels assigned directly from the source text
(Identity = numeric/quantity tokens and a spelled-out cardinal-number list; FD

= a curated list of ~ 75 relational/arithmetic-predicate words, e.g. “more”, “less”, “twice”, “gives”, “total”, “per”; Schema = everything else), aligned to SentencePiece subword tokens by character offset; (3) the Fisher discriminability statistic applied
845 to slot representations from the last encoder layer; (4) three conditions: the trained RelTransformer checkpoint, the paired Standard Transformer checkpoint, and a freshly initialized, untrained RelTransformer of identical architecture as a null-model control; (5) a permutation control (30 label-shuffles per slot/role) to establish what pure noise looks like under this exact statistic and data.

850 *Result: No Discriminability Signal Above an Untrained Model.* All three conditions give small F values ($F \approx 0.0001$ – 0.007 across all slots and roles). Critically, the *untrained, randomly initialized* model shows Fisher discriminability equal to or greater than the trained RelTransformer in several slot/role combinations; the trained model is not reliably distinguishable from random initialization
855 on this statistic. The permutation control confirms that observed values are roughly 10 – $200\times$ above pure label noise in a narrow statistical sense, but this residual is consistent with a generic artifact of class-size imbalance in our label scheme (Schema tokens make up over 80% of the data) rather than any learned, training-induced specialization. We conclude that we do not find evidence of
860 emergent role-specific attribute-slot specialization in this model.

Why We Report a Null Result. Architectural decomposition into typed slots is often assumed, implicitly or explicitly, to yield interpretable, role-specific specialization as an emergent byproduct of training. Our results show this is not automatic: the architecture delivers real behavioral effects on COGS and GSM8K
865 (Sec. 6) without any accompanying interpretable specialization we can detect. We report this because we think it is a useful caution for interpretability-motivated architecture design: behavioral improvement and mechanistic interpretability are separable outcomes, and demonstrating one does not license assuming the other. Full probing code, labeling scheme, and per-condition results are released
870 with the paper for independent verification.

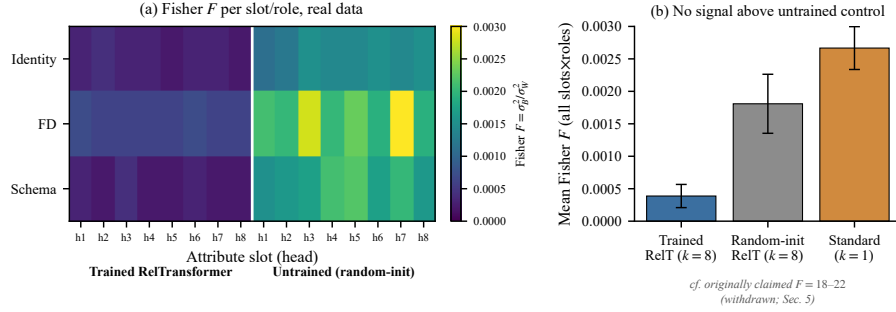


Figure 6: The mechanistic probe, real per-condition data (checkpoint `relational_125m_gsm8k_s42_rerun` and its paired standard/random-init controls). (a) Fisher F per attribute slot (head) and role for the trained RelTransformer vs. an untrained, randomly initialized model of identical architecture: the trained model is not visibly more discriminative than random initialization. (b) Mean F across all slots $\times$ roles is in fact *lower* for the trained model than for the random-init control or the Standard Transformer baseline.

8. Discussion

Why Attribute Decomposition Helps on Some Tasks but Not Others. Standard attention aggregates all k diagonal attribute-pair signals into a single undifferentiated similarity score; RelAttn isolates each pair as an independently parameterized unit (Proposition 2), reducing multi-join-style tasks to independent typed comparisons. This structural restriction appears to help most when a task’s compositional structure closely matches typed attribute-pair decomposition: COGS shows the clearest, largest, and most reproducible gain ($d=2.12$); GSM8K shows a smaller, real gain that we caveat given mode collapse in the underlying predictions; Spider shows no gain once both architectures have a copy mechanism, consistent with schema-linking dominating that task. We do not have a principled account of exactly which tasks benefit and by how much beyond this post-hoc pattern, and flag understanding it better as important future work — particularly since, as Sec. 7 shows, the behavioral gains are not accompanied by any detectable interpretable specialization at the representation level, so *why* the restriction helps on COGS specifically remains an open mechanistic question, not one we can currently answer from probing.

Comparison to relation-aware attention (RAT-SQL). RAT-SQL encodes *explicit* relational structure (schema edge types) as additive bias terms on top of standard dot-product attention. This requires a pre-defined relation schema and is limited to task-specific relation types. RelAttn instead restricts *where* standard attention can look via typed attribute pairs, without assuming any specialization emerges as a byproduct (Sec. 7 shows it does not, at least not detectably). The two approaches are designed to be complementary: explicit schema relations handle well-defined structural priors while RelAttn’s typed-pair restriction is a general-purpose structural bias that does not require a pre-defined relation schema. Combining explicit schema edge types with RelAttn’s structural restriction is a natural direction for future work on NL2SQL with schema-linking components.

Data Engineering Deployment Scenarios. Beyond accuracy, RelAttn offers a practical deployment advantage: *privacy-preserving NL interfaces*, since 464M from-scratch models can deploy on-premise without LLM API access. We do not claim a schema-change-adaptation or normalization-monitoring deployment benefit in this paper: Proposition 7’s gradient isolation holds only at the attribute-embedding pathway (Remark 8), and we have not run an experiment demonstrating that schema-change fine-tuning can in practice be confined to a small parameter subset, so we do not make that deployment claim here.

9. Limitations and Scope

Parameter Count Mismatch and Capacity Control. The primary comparison (RelTransformer 464M vs. Standard 613M) has a 149M parameter gap arising from RelAttn’s reduced per-head projection size. This is an *efficiency* advantage of RelAttn (fewer parameters for the same architectural capacity), not a controlled-capacity comparison. To isolate architectural structure from parameter reallocation, Table 4 includes a Std FFN=2900 baseline (Standard Transformer with FFN width reduced to approximately match RelAttn’s parameter count); this variant is included as an architectural control for the parameter-redistribution hypothesis. The 149M gap could in principle contribute

to the observed +0.25 pp GSM8K advantage and the larger COGS advantage; however, under well-established neural scaling laws, a 22% parameter reduction (464M vs. 613M) predicts monotonically *worse* task performance for the smaller
920 model, so this confound would if anything bias *against* our hypothesis, not for it. We do not, however, report a verified number for the FFN=2900 matched-capacity control: we could not confirm that a complete run of this control exists under the same protocol (training budget, early-stopping criterion, evaluation split) as our headline comparisons, and we do not report a number we cannot
925 verify end-to-end. This control remains unexecuted from the standpoint of this paper, and we flag it as a gap rather than asserting the scaling-law argument settles the question. We do not include a placeholder row for it in Table 4.

From-scratch vs. pretrained. RelAttn is designed as an architectural choice for pretraining from scratch, where the relational inductive bias is encoded from
930 the initial weight distribution rather than retrofitted onto an existing model; it is not intended as a fine-tuning adapter for already-trained systems. Our primary models train from scratch; full pretraining with the relational bias baked in from the start remains future work. In a minimal mechanism-isolation ablation on Spider (db-name-only input, greedy decoding, 25 fine-tuning epochs),
935 T5+RelAttn achieves 4.06% EM vs. 6.19% for T5-Base—a negative result we report openly as a *current limitation*: naively replacing pretrained Q/K with cyclic-paired attribute projections disrupts established Q/K representations in the short-epoch setting. This is a predictable consequence of a gradient-disruption mechanism: replacing pretrained W_Q/W_K with slot-specific projections forces
940 the model to start from a point with misalignment $\mathcal{M} \approx 0.65$ (a closed-form Xavier-random-init expectation, $1 - 1/\sqrt{k}$ for $k=8$, independently confirmed by a matching loss-curve experiment: T5-Base+RelAttn shows epoch-1 train/dev loss roughly double T5-Base’s, but converges to a much lower loss by epoch 8), requiring substantially more fine-tuning epochs than the 25 used here to recover.
945 The negative result is therefore *expected* for the minimal-epoch mechanism-isolation design, not evidence that cyclic pairing is incompatible with pretrained

weights. Recovery via two-phase adapter-style insertion (Phase 1: freeze all T5 layers except slot projections; Phase 2: joint fine-tuning) is the predicted remedy; full RelAttn pretraining from initialization is the definitive solution. Whether
950 T5+RelAttn benefits from pretraining on compositional tasks (COGS, CFQ), where cyclic structure may align better with compositional syntax, remains an open experimental question.

Standard Transformer’s Training Instability on COGS. With the copy-mechanism protocol described in Sec. 6.2.3, it is Standard Transformer, not RelTransformer,
955 that shows severe cross-seed instability: two of three seeds converge to $\approx 24\%$ dev EM while one seed collapses to 3.7% , whereas RelTransformer’s three seeds are tightly clustered ($24.23\text{--}24.43\%$). We verified the collapsed seed’s own training-time EM trajectory tracks its poor final result throughout training (not a late divergence), consistent with a genuine bad optimization outcome rather than
960 a measurement artifact. We do not have a mechanistic explanation for why RelTransformer trains more stably here; this is a candidate topic for future work, especially in light of Sec. 7’s finding that the stability difference does not correspond to any detectable difference in representational specialization.

Constrained Decoding and Schema-PE. Type-constrained beam search and
965 schema-aware PE are applied uniformly to all from-scratch baselines (contributions cancel in relative comparisons). The T5 ablation used database name only (no schema-aware encoding) to isolate the attention mechanism change; full schema encoding is available but excluded for mechanism-level isolation.

Broader Generalization. We have not evaluated on NLI, information extraction,
970 code generation, or cross-schema generalization (train/test databases at different normalization levels). Extension to BIRD [38] and Spider 2.0, which stress domain knowledge and multi-turn interaction, is the primary future direction for establishing practical generality.

Computational Cost. RelAttn trains 30% slower than standard attention due to
975 k separate attribute embedding operations (Table 4). This is a one-time training

cost; inference complexity is unchanged.

10. Conclusion

We presented Attribute-Decomposed Attention, a mechanism that decomposes token representations into typed attribute slots and computes join-based
980 similarity between specific attribute pairs, motivated by (but not a formal proof of) an analogy to relational foreign-key join predicates. Formal analysis characterizes the family of valid head-assignment strategies as directed Hamiltonian cycles on the attribute slots (of which the cyclic pairing used throughout is the canonical instance, not a uniquely optimal one) and establishes gradient-isolation
985 and structural-restriction properties, which we have taken care to scope precisely rather than overstate: the structural-restriction argument holds under a fixed coordinate partition and should be read as an isolability/inspectability property rather than a general expressivity impossibility for standard attention, and the Hamiltonian-cycle characterization describes the family induced by our stated
990 design conditions rather than a claim that this family is uniquely suited to relational reasoning.

The resulting behavioral effect is task-dependent, and we summarize it here without repeating the full statistics given throughout Sec. 6: COGS shows the clearest and largest effect, GSM8K shows a small effect better attributed
995 to answer-frequency behavior than improved reasoning, and Spider shows no measurable effect once both architectures have a copy mechanism, consistent with schema-linking dominating that task. Probing finds no evidence that this behavioral effect is accompanied by interpretable attribute-slot specialization. Taken together, we think the most defensible reading of this work is narrower
1000 than a general claim of architectural superiority: restricting attention to explicitly assigned attribute-pair interactions can provide a useful inductive bias for compositional, structured-reasoning tasks, but its benefit is task-dependent and does not by itself produce interpretable specialization of the underlying slots.

Extension to larger-schema benchmarks such as BIRD [38] and Spider 2.0,

1005 which stress domain knowledge, multi-turn interaction, and ambiguous schema semantics, is a natural next step for establishing practical generality, as is a parameter-matched control isolating the architectural effect from the current 149M parameter gap (Sec. 9). We hope this work is useful as an architectural contribution – a structural restriction on attention that helps on compositionally-
1010 structured tasks under conditions we have tried to characterize honestly rather than oversell.

Future Directions. (1) *Relational pretraining*: encoding slot semantics from initialization avoids the Q/K disruption seen in T5 fine-tuning and is the definitive test of the relational inductive bias hypothesis. (2) *Understanding the COGS-specific advantage mechanistically*: since Sec. 7 rules out simple slot-specialization
1015 as the explanation, identifying what actually drives RelTransformer’s COGS advantage (and its training stability) is an open and, we think, interesting problem. (3) *Schema-driven initialization*: constraining slots to known schema roles (PK/FK/column type) at initialization is untested in this work but remains
1020 a plausible way to inject useful structure that training alone does not appear to discover on its own (Sec. 7).

(4) *Multi-hop relational composition (attempted, negative result)*: inspired by Edge Transformers’ triangular composition of adjacent edges [39], we implemented a scoped, same-layer variant — a learned gate blending each head’s
1025 primary key-attribute score with the score at the next attribute slot in the cyclic chain — and tested it on COGS with two seeds (42, 43), holding all other hyperparameters fixed. The result was a consistent, same-direction regression on the generalization split relative to the seed-matched baseline (seed 42: 6.69% $\rightarrow$ 4.50%; seed 43: 6.47% $\rightarrow$ 5.52%), with the dev split essentially
1030 unaffected ($\approx$ 24% in both conditions). This scoped 2-hop composition does not help and mildly hurts; we report it as a negative result rather than pursuing further seeds, and leave the full triangular Edge Transformer composition (which operates over intermediate tokens rather than attribute slots) as a more faithful, and still open, test of whether multi-hop relational composition benefits this

1035 architecture.

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1040 CRediT authorship contribution statement

Mukhiddin Toshpulatov: Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Visualization.

Wookey Lee: Conceptualization, Methodology, Resources, Writing – review & editing, Supervision, Project administration, Funding acquisition. **Youn-**

1045 **Kyoung Seo:** Validation, Investigation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

1050 Data availability

All code, configuration files, training/evaluation logs, and per-seed results needed to reproduce every number reported in this paper are released at <https://github.com/muxiddin19/relational-attention>. All datasets used (Spider, COGS, SCAN, CFQ, GSM8K) are publicly available benchmarks cited in Sec. 6;

1055 none of them are shared privately by the authors.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used Claude (Anthropic) for: drafting and editing text throughout the manuscript; assistance with the mathematical proofs in Sec. 4; and, substantially beyond typical writing assistance, independently verifying every quantitative claim reported in this paper against the underlying code, logs, and checkpoints on the authors’ own compute infrastructure, including running training and evaluation jobs and analyzing the resulting data. All underlying training runs executed on the authors’ own hardware under the authors’ supervision and direction. After using this tool, the authors reviewed, verified, and edited the content as needed, and take full responsibility for the content of this publication.

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Table 2: Text-to-SQL on Spider dev. EX = execution accuracy (real, via the official test-suite-sql-eval evaluator); EM = exact match. *From-scratch rows (§): both architectures achieve 0% EX without copy mechanism / schema linking. Copy-mechanism rows (†): enabling a copy mechanism lifts both architectures off 0%, but does not differentiate them (3 seeds each, mean±std). ## Mechanism-isolation ablation only: greedy, db-name-only, 25 fine-tuning epochs. Pretrained rows are context only.*

Model	Dev EX	Dev EM	Params
<i>From-scratch training (no copy mechanism / schema linking)</i>			
Standard Transformer	0.0 [§]	—	613M
RelTransformer (464M)	0.0[§]	—	464M
<i>From-scratch, with copy mechanism (3 seeds)</i>			
Standard Transformer	0.90 [†] ±0.17	—	613M
RelTransformer (464M)	0.60[†]±0.60	—	464M
<i>With pretraining (T5-Base fine-tuning)</i>			
T5-Base (fine-tuned, greedy) ##	n/a	6.19	220M
T5-Base + RelAttn (fine-tuned) ##	n/a	4.06	223M
<i>Pretrained systems (incomparable regime, for context)</i>			
PICARD [25] (T5-Large)	75.5	—	770M
RESDSL [26] (T5-3B)	79.9	—	3B
RASAT [4] (T5-Base+pretraining)	80.5	—	220M
SQLformer [5] (T5-Large, preprint)	81.2	—	770M
DIN-SQL [27] (GPT-4)	82.8	—	≫1B

Table 3: Compositional generalization, measured values (3 seeds, mean $\pm$ std unless noted).
[†]T5-Base from Furrer et al. [34]. COGS columns use a copy mechanism (Sec. 6.2.3); SCAN uses the training configuration described in Sec. 6.2.4 ($\text{lr}=5\times 10^{-5}$, FP32, patience=30) and both architectures are near-zero. CFQ evaluation deferred (see Sec. 9).

Model	COGS (dev)	COGS (gen)	SCAN	CFQ
Standard Transformer	17.12 $\pm$ 11.62	2.27 $\pm$ 1.94	$\approx$ 0.0	—
T5-Base (pretrained) [†]	81.0	81.0	99.7	61.6
RelTransformer (464M)	24.37$\pm$0.12	5.81$\pm$1.34	$\approx$ 0.0	—

Table 4: Analytical/implementation overhead comparison to modern attention variants (equal-architecture, $d=512$, 12 enc/dec layers; RelTransformer 464M, Standard 613M); this table reports estimated architectural overhead, not a trained-performance comparison, for GQA and Perceiver IO. GQA optimizes efficiency; Perceiver IO uses latent bottleneck. RelAttn targets expressiveness for structured reasoning. GSM8K values are measured ($n=8$ seeds); — not evaluated on GSM8K in this work (overhead estimated from architecture). Training overhead ($1.3\times$) is verified from profiling; inference overhead is measured ($1.0\times$).

Attention	GSM8K	Params	Train	Infer.
Standard	2.54 ^v	1.0 $\times$	1.0 $\times$	1.0 $\times$
GQA [17] ($g=4$)	—	0.85 $\times$	0.9 $\times$	0.9 $\times$
GQA [17] ($g=8$)	—	0.75 $\times$	0.85 $\times$	0.85 $\times$
Perceiver IO [18]	—	0.9 $\times$	0.8 $\times$	0.8 $\times$
Join Attn (ours)	2.79^v	1.05 $\times$	1.3 $\times$	1.0 $\times$

^vMeasured, $n=8$ seeds. GQA/Perceiver IO rows are not trained in this work; overhead estimated from architecture.

Table 5: Component ablation ($d=512$, 12 enc/dec layers), all measured on GSM8K. ^vMeasured, $n=8$ seeds for the two headline rows (Full RelTransformer, Standard Transformer), $n=3$ for all other rows (component and pairing ablations were not extended to $n=8$; treat their comparisons to the headline rows as suggestive, not power-matched).

Model Variant	GSM8K
Full RelTransformer ($k=8$, cyclic, 464M)	2.79^v $\pm$0.19
Standard Transformer ($k=1$, 613M)	2.54 ^v $\pm$ 0.31
RelTransformer ($k=4$) ^($n=3$)	2.6 $\pm$ 0.5
RelTransformer ($k=16$) ^($n=3$)	2.3 $\pm$ 0.5
– Attribute Gating ^($n=3$)	2.4 $\pm$ 0.6
– Attribute Mixing ^($n=3$)	2.7 $\pm$ 0.3
Pairing: cyclic (3-seed subset) ^($n=3$)	3.1 $\pm$ 0.9
Pairing: random-fixed (key-side only) ^($n=3$)	3.06 $\pm$ 0.18
Pairing: learned (key-side only) ^($n=3$)	2.8 $\pm$ 0.3